

Multiplying the load with Z-factor factor gives you the value of battery backup required for one hour. Let's say you have a 90W load. So for one hour, the battery backup is: $90 \times Z = 19.998 \text{ Ah} \approx 20 \text{ Ah}$. For six hours, the battery backup required is $90 \times 6Z = 119.98 \text{ Ah} \approx 120 \text{ Ah}$.

Limitations of ZMR backup analysis are:

1. The performance of Z-factor decreases with aging effect of batteries.
2. It supports a load upto 5kW.

If you know the current consumption of the load, it is simple to calculate the Ah for the required load by formula: $\text{Ah} = I \times T$

where 'I' is current consumption of load in amps and 'T' is backup time in hours. For example, if $I = 10 \text{ A}$ and $T = 3 \text{ hours}$, $\text{Ah} = 10 \times 3 = 30 \text{ Ah}$.

As you can see from Table II, a 400W inverter with 12V, 120Ah battery is unable to support a computer with UPS. After two minutes, the voltage drops.

When the UPS is removed and a voltage regulator with 230V output placed, the 400W inverter is able to run the computer without any interruption. The supply to the computer without UPS is shown in Table III. It continues for up to six hours, then shuts down.

New ideas

Given below are some new ideas for making use of solar energy:

Solar batteries. Solar batteries are 12V DC batteries with different Ah

The battery backup for the required load can be calculated by Z-factor, where $Z=0.2222$. This Z-factor has been found out from the observations with batteries used for certain loads. As the wattage increases, this 'Z' value also increases

TABLE IV
Advantages and Disadvantages of Various Power Plants

	Advantages	Disadvantages
Thermal power plant	Low cost, high efficiency	Pollution, non-renewable energy, very expensive, waste products are harmful
Nuclear power plant	Low cost, high efficiency	Harmful radiations are produced. The plants are costlier to establish. Nuclear wastes are harmful to human beings, environment and animals. It is difficult to stop nuclear fusion once it starts. Radiation leakage could lead to disasters.
Solar power plant	More energy can be produced at lower cost. The plants can be established in desert areas. These are eco-friendly as there is no pollution. Also, the energy is renewable.	High initial cost to establish the power plant, less efficient than other energy sources.

capacities. These batteries have 12V solar panels on top of them beside, of course, the charging circuitry. You just have to place the battery in the sun for charging and use it after it is completely charged. A heat-insulating material is coated to prevent the environmental heat from entering the battery, thus protecting the battery from damage due to over-heating.

Solar tanks. Just like a water tank on top of the buildings, we can have solar tanks that absorb the sunlight and supply power to the houses for a long time. These can be built in the same way as a solar battery but have to be bigger in size.

Nano solar cells. As the power consumption increases, the required wattage of the solar panel also increases, requiring an increase in its size. Nanotechnology allows reduction in size of the solar cells while increasing their efficiency.

Solar nets. With computers running on solar energy, there would be no need to pay electricity bill for the computer usage. A package of computer with the solar panel, solar converter and battery would make this possible.

Solar air-conditioners. A conventional split air-conditioner has the condenser installed outside the house and conditioning unit placed inside a room. If the outer body is surrounded by walls of solar panels, it's possible to run the air-conditioner with solar en-

ergy with the help of a small converter and battery.

Solar power generators. The electric generators work on fuel. Due to the combustion of fuel, a lot of gases including CO_2 and CO are released into the atmosphere. By replacing the fuel with solar energy, it is possible to generate electricity without any pollution. For example, consider a 5kVA generator. A motor is used to run the generator in order to produce 5kVA supply. The energy required to rotate this motor can come from a solar panel, whose DC supply is converted into AC using a solar inverter.

Solar robots. A robot can be made to work using DC supply for all its PCB components including the microcontroller. For this, the robot can be made with DC components and its outer body covered with solar cells. The solar energy will be directly absorbed by the robot; the robot will recharge when in sunlight and use the stored energy.

Solar laptop. Solar panels can be placed on the body of a laptop to charge the battery as well as use the supply. There can also be the option to use electrical supply when sunlight is unavailable because of the monsoon or night time. ●

P. Rajesh is an M.Tech and assistant professor, N. Pavan Kumar is MA, M.Phil and assistant professor, and Syed Zameeruddin is a third-year B.Tech student